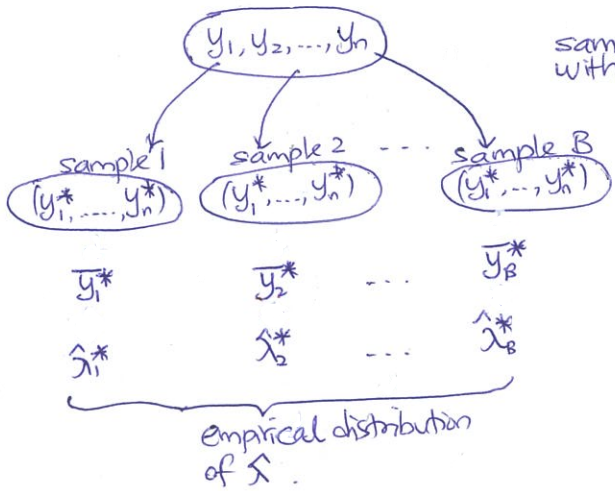
FAMB: Chapter 4

Bootstrap

Suppose we have a random sample $y_1, y_2, \ldots, y_n$ with $y_i \sim \text{Exp}(\lambda)$

Nonparametric bootstrap

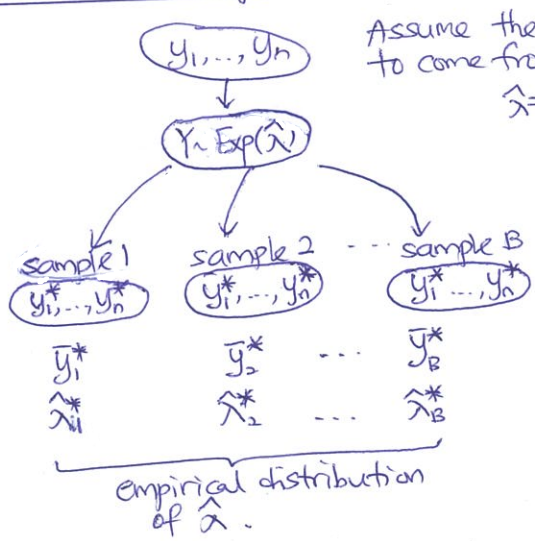


$$\hat{E}(\hat{\lambda}) = \frac{1}{B}\sum_{j=1}^{B}\hat{\lambda}_j^*$$

$$\widehat{Var}(\hat{\lambda}) = \frac{1}{B-1}\sum_{j=1}^{B}(\hat{\lambda}_j^* - \bar{\lambda}^*)^2 \quad \text{where} \quad \bar{\lambda}^* = \frac{1}{B}\sum_{j=1}^{B}\hat{\lambda}_j^*$$

N/t: The C.I is usually referred to as the percentile bootstrap C.I.

Parametric bootstrap



$$\hat{E}(\hat{\lambda}) = \frac{1}{B}\sum_{j=1}^{B}\hat{\lambda}_j^*$$

$$\widehat{Var}(\hat{\lambda}) = \frac{1}{B-1}\sum_{j=1}^{B}(\hat{\lambda}_j^* - \bar{\lambda}^*)^2 \quad \text{where} \quad \bar{\lambda}^* = \frac{1}{B}\sum_{j=1}^{B}\hat{\lambda}_j^*$$

In parametric boostrap, instead of simulating bootstrap samples that are iid from the empirical distribution (the non-parametric estimate of the distribution of the data), we simulate bootstraps samples that are iid from the estimated parametric model.